



SRI RAMAKRISHNA ENGINEERING COLLEGE

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MINI-PROJECT REPORT

SMART CHEMICAL LAB MONITORING SYSTEM

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In partial fulfillment of the course

25CS2102 – SENSORS FOR ENGINEERING APPLICATION

COMPUTER SCIENCE AND ENGINEERING

SRI RAMAKRISHNA ENGINEERING COLLEGE

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Vattamalaipalayam, N.G.G.O. Colony Post,

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SMART CHEMICAL LAB MONITORING SYSTEM

INTRODUCTION:

The Smart Chemical Gas Monitoring System is an IoT-based safety solution designed to detect harmful gases, monitor room temperature, and sense light conditions in real time. This system is highly useful in chemical laboratories, industries, and smart homes where safety and environmental monitoring are critical.

Traditional monitoring systems require manual checking, which can be inefficient and unsafe. This project uses sensors and a microcontroller to automatically detect gas leaks, temperature changes, and light intensity. When abnormal conditions are detected, the system alerts users through a buzzer and displays data on an LCD. It can also send data online for remote monitoring.

COMPONENTS USED :

1. MQ 2(gas sensor)

Gas sensors are devices used to detect the presence and concentration of gases in the environment. They play a critical role in safety systems, industrial monitoring, and environmental protection. In this project, MQ-series gas sensors are used due to their low cost, high sensitivity, and ease of interfacing with microcontrollers.

MQ gas sensors work based on the chemiresistive sensing principle. The sensor contains a sensing material (usually tin dioxide – SnO_2) whose electrical resistance changes when exposed to gases.

- In clean air → high resistance
- In presence of gas → resistance decreases

The sensor outputs an analog voltage based on gas concentration.



Use in the project: The MQ 2 gas sensor is useful in detection of gas in the chemical lab environment.

2. DHT 22(Temperature sensors):

Temperature sensors measure thermal energy in the environment. The DHT11 is widely used due to its simplicity and digital output.

The DHT11 uses a **thermistor** to measure temperature.

- Resistance changes with temperature
- Internal ADC converts analog signal to digital



Use in the project: This sensor is used to get temperature using thermal energy present in the environment.

3. LIGHT SENSORS(LDR):

A Light Dependent Resistor (LDR) is used to measure light intensity. LDR works on **photoconductivity**:

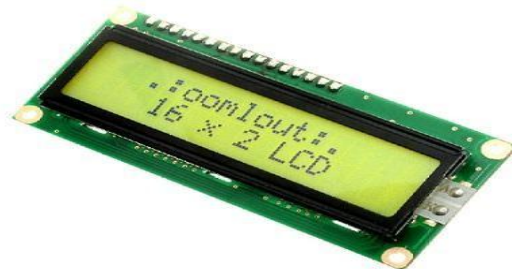
- More light → less resistance
- Less light → high resistance



Use in the project: The LDR used to detect the light in the environment.

4. LCD Display

The **16x2 LCD display** is used to show system status messages. It operates using the **I2C protocol** or a **parallel interface** and is controlled by the microcontroller. The display provides feedback to users by showing messages such as "GAS LEAK", "EVACUATE LAB", "TEMP HIGH", "LOW LIGHT".



Use in the project: . The display provides feedback to users by showing messages such as "GAS LEAK", "EVACUATE LAB", "TEMP HIGH", "LOW LIGHT".

5. Buzzer

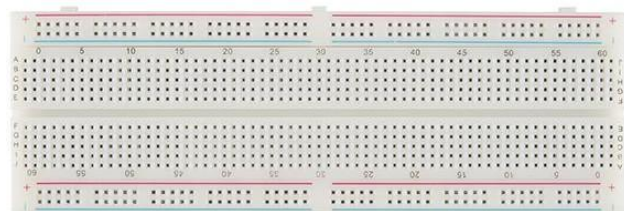
The buzzer acts as an **alarm device** that notifies users immediately when sensor readings cross safe limits. Unlike visual alerts (LEDs or screens), a buzzer ensures attention even if no one is actively monitoring the system.



Use in the project: The buzzer beeps when the gas sensor detects above 200.

6. Breadboard & Jumper Wires

A **breadboard** is a reusable platform for building and testing electronic circuits without soldering. It consists of multiple rows of **metal strips** that form electrical connections when components are inserted. **Jumper wires** are used to establish electrical connections between different components.



Use in the project: The breadboard and jumper wires help in **prototyping** the circuit by connecting all components securely and allowing easy modifications.

WORKING:

The working principle of a smart chemical lab monitoring system is based on continuous sensing, data processing, and automatic alerting to ensure safety in the laboratory environment. It integrates multiple sensors, a controller, and alert mechanisms to detect hazardous conditions in real time.

1. Data Sensing (Input Stage)

The system uses different sensors to monitor environmental conditions:

- Gas sensors (e.g., MQ-2 Gas Sensor)
Detect harmful gases like LPG, methane, or smoke.
- Temperature & humidity sensors (e.g., DHT11 Sensor)
Measure temperature changes that may indicate overheating or fire risk.
- Light sensors (LDR)
Detect light intensity to monitor lab activity or unexpected conditions.

These sensors continuously collect data from the surroundings.

2. Data Processing (Control Unit)

All sensor data is sent to a microcontroller such as the Arduino Uno.

- The controller reads sensor values at regular intervals.
- It compares the values with predefined safety thresholds.
- Logical decisions are made based on conditions programmed into the system.

3. Decision & Alert Mechanism

If any parameter exceeds safe limits:

- A buzzer is activated for audible alerts
- LEDs may glow as visual indicators
- Notifications can be sent (if IoT-enabled system is used)

Example:

- Gas level high → buzzer ON
- Temperature too high → alarm triggered
- Normal conditions → system remains idle

4. Optional IoT Integration

In advanced systems, the monitoring unit is connected to the internet:

- Data is sent to cloud platforms
- Users can monitor conditions remotely via mobile apps
- Alerts (SMS/email) are sent instantly

5. Continuous Monitoring Loop

The system operates in a loop:

1. Sense data
2. Process data
3. Compare with thresholds
4. Trigger alerts if needed
5. Repeat continuously

PROGRAM :

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

// ===== DHT22 Setup =====
#define DHTPIN 2
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

// ===== Sensor Pins =====
#define GAS_SENSOR A0
#define LDR_SENSOR A1

// ===== Output Pins =====
#define GAS_LED 6
#define TEMP_LED 7
#define BUZZER 8

// ===== LCD Setup =====
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {

    Serial.begin(9600);

    dht.begin();

    pinMode(GAS_LED, OUTPUT);
    pinMode(TEMP_LED, OUTPUT);
    pinMode(BUZZER, OUTPUT);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0,0);
    lcd.print("CHEM LAB");
    lcd.setCursor(0,1);
    lcd.print("MONITORING...");
    delay(2000);
    lcd.clear();
}

void loop() {

    float temp = dht.readTemperature();
    float hum = dht.readHumidity();

    int gasValue = analogRead(GAS_SENSOR);
    int lightValue = analogRead(LDR_SENSOR);

    // ===== Display Normal Data =====
```



```

lcd.setCursor(0,0);
lcd.print("T:");
lcd.print(temp);
lcd.print("C ");

lcd.print("H:");
lcd.print(hum);

lcd.setCursor(0,1);
lcd.print("Gas:");
lcd.print(gasValue);

delay(1500);

// ===== GAS ALERT =====
if(gasValue > 400) {

    digitalWrite(GAS_LED, HIGH);
    digitalWrite(BUZZER, HIGH);

    lcd.clear();
    lcd.print("GAS LEAK!");
    lcd.setCursor(0,1);
    lcd.print("Evacuate Lab");

    delay(2000);
}
else {
    digitalWrite(GAS_LED, LOW);
    digitalWrite(BUZZER, LOW);
}

// ===== HIGH TEMPERATURE ALERT =====
if(temp > 45) {

    digitalWrite(TEMP_LED, HIGH);
    digitalWrite(BUZZER, HIGH);

    lcd.clear();
    lcd.print("TEMP HIGH!");
    lcd.setCursor(0,1);
    lcd.print("Cooling Req.");

    delay(2000);
}
else {
    digitalWrite(TEMP_LED, LOW);
}

// ===== LOW LIGHT ALERT =====
if(lightValue > 800) { // Dark condition

    digitalWrite(BUZZER, HIGH);

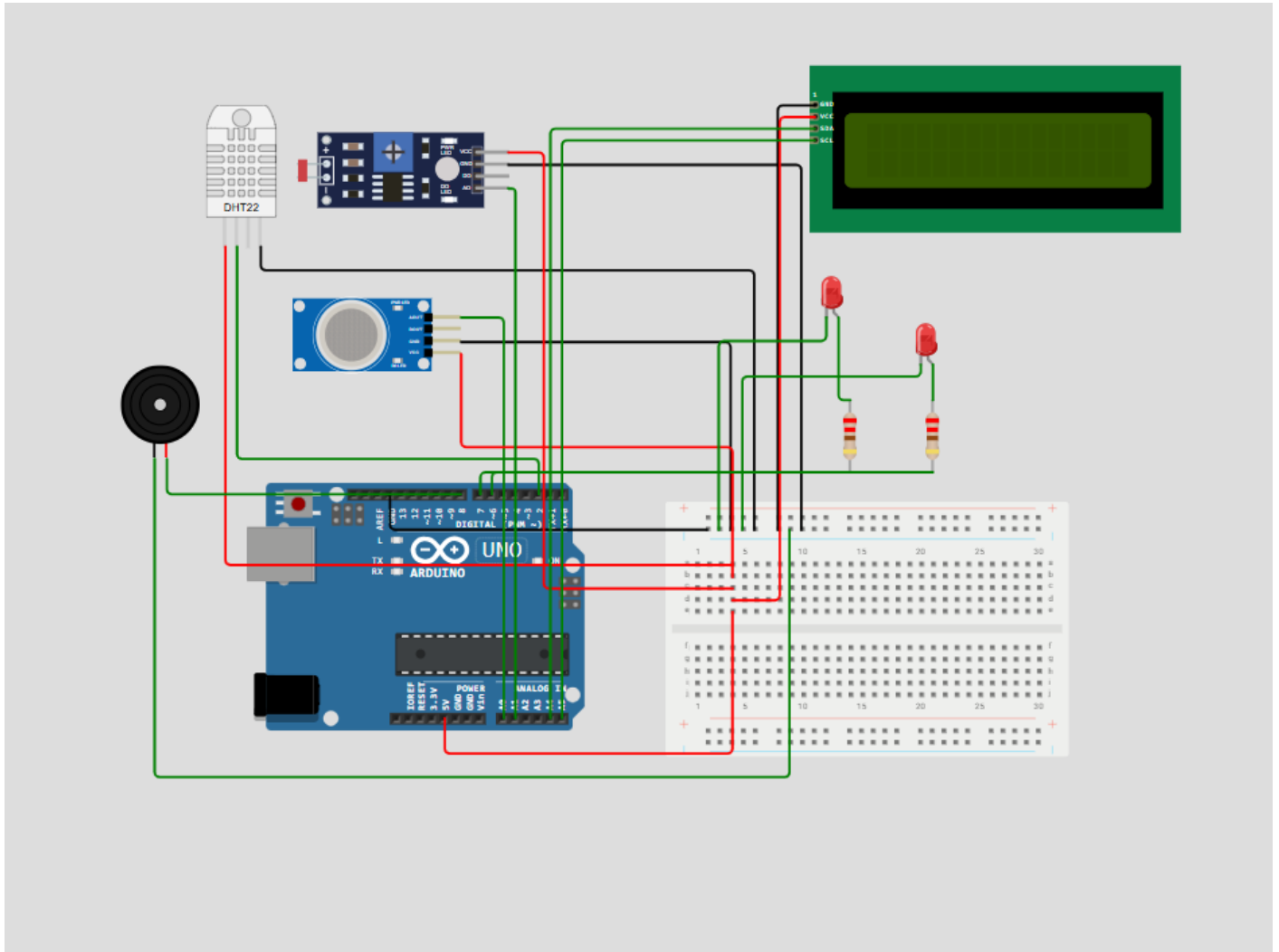
    lcd.clear();
    lcd.print("LOW LIGHT!");
    lcd.setCursor(0,1);

```

```
    delay(2000);  
}  
  
digitalWrite(BUZZER, LOW);  
  
lcd.clear();  
}
```

CIRCUIT DIAGRAM :

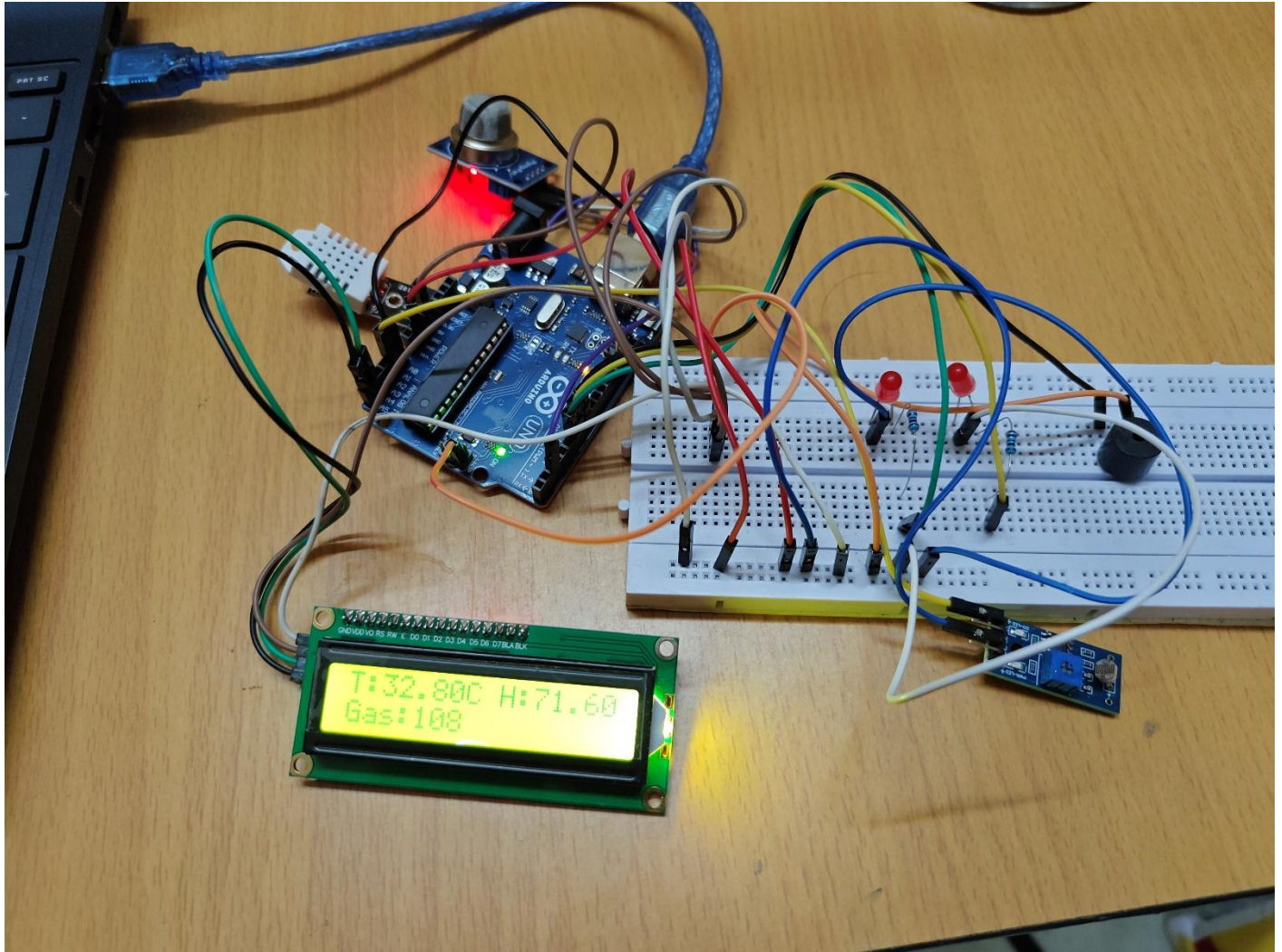
The circuit includes sensors, a microcontroller, and output devices. The Arduino Uno controls the entire system. Sensors like MQ-2 Gas Sensor and DHT11 Sensor collect environmental data. The controller processes the data and checks for unsafe conditions. If limits are exceeded, the buzzer and LED provide alerts.



RESULTS:

The system successfully detected harmful gases, monitored temperature, and sensed light intensity in real time. The gas sensor triggered the buzzer when unsafe levels were detected, ensuring quick alerts. Temperature and light readings were accurately displayed on the LCD. Overall, the system showed reliable performance, fast response, and effective integration of all sensors.

WORKING MODEL:



CONCLUSION:

The Smart Chemical Gas Monitoring System with Light and Temperature Detection was successfully implemented and tested under different environmental conditions. The system performed effectively in detecting gas leakage, measuring temperature, and sensing light intensity in real time.

During testing, the gas sensor responded quickly to the presence of gases such as LPG and smoke. When the gas concentration exceeded the predefined threshold value, the system immediately activated the buzzer, providing an audible alert. This confirms the system's ability to detect hazardous conditions and respond promptly to ensure safety.

The temperature sensor accurately measured the ambient temperature and displayed the values on the LCD screen. When the temperature crossed the safe limit, the system triggered a warning through the buzzer, indicating abnormal conditions. The readings were consistent and reliable under normal operating conditions.

The light sensor (LDR) successfully detected variations in light intensity. It was able to differentiate between bright and dark environments, and the corresponding status was displayed on the LCD. This demonstrates the system's capability to monitor environmental lighting effectively.

The LCD display continuously showed real-time values of gas levels, temperature, and light intensity, providing a user-friendly interface for monitoring. The buzzer worked efficiently as an alert mechanism, activating instantly during unsafe conditions.

Overall, the system showed stable performance, quick response time, and accurate sensing capabilities. The integration of multiple sensors worked seamlessly, proving the effectiveness of the design. The project successfully meets its objective of providing a low-cost, reliable, and efficient environmental monitoring and safety system.